

# API Design for Option Pricing Application

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## Overview

This document outlines the REST API design for the option pricing application that compares Pareto (Power Law), Black-Scholes-Merton, and market prices for tail options. The API serves 100 traders with approximately 2,000 requests per day.

## Base URL

```
Production: https://api.optionpricing.com/v1
Development: http://localhost:8000/v1
```

## Authentication

All API endpoints require Bearer token authentication.

### Authentication Header

```
Authorization: Bearer <jwt_token>
```

### Get Access Token

**POST** /auth/token

#### Request Body:

```
{
  "username": "trader001",
  "password": "secure_password"
}
```

#### Response:

```
{
  "access_token": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9...",
  "token_type": "bearer",
  "expires_in": 3600,
  "trader_id": "550e8400-e29b-41d4-a716-446655440000"
}
```

# API Endpoints

## 1. Trader Management

### Get Trader Profile

**GET** /traders/profile

#### Headers:

```
Authorization: Bearer <jwt_token>
Content-Type: application/json
```

#### Response:

```
{
  "trader_id": "550e8400-e29b-41d4-a716-446655440000",
  "username": "trader001",
  "email": "trader001@firm.com",
  "full_name": "John Smith",
  "organization": "ABC Trading",
  "created_at": "2024-01-01T00:00:00Z",
  "last_login": "2024-01-15T10:30:00Z",
  "is_active": true
}
```

### Update Trader Profile

**PUT** /traders/profile

#### Request Body:

```
{
  "full_name": "John A. Smith",
  "organization": "ABC Trading LLC"
}
```

#### Response:

```
{
  "message": "Profile updated successfully",
  "trader_id": "550e8400-e29b-41d4-a716-446655440000"
}
```

## 2. Market Data

### Get Current Market Data

**GET** /market-data/{symbol}

#### Example Request:

```
GET /market-data/SPY
Authorization: Bearer <jwt_token>
```

#### Response:

```
{
  "market_data_id": "660e8400-e29b-41d4-a716-446655440001",
  "symbol": "SPY",
  "spot_price": 415.50,
  "risk_free_rate": 0.0525,
  "data_date": "2024-01-15",
  "created_at": "2024-01-15T09:30:00Z"
}
```

### Get Market Data History

**GET** /market-data/{symbol}/history

#### Query Parameters:

- **start\_date** (optional): YYYY-MM-DD format
- **end\_date** (optional): YYYY-MM-DD format
- **limit** (optional): Number of records (default: 30)

#### Example Request:

```
GET /market-data/SPY/history?start_date=2024-01-01&end_date=2024-01-15&limit=10
Authorization: Bearer <jwt_token>
```

#### Response:

```
{
  "symbol": "SPY",
  "data": [
    {
      "market_data_id": "660e8400-e29b-41d4-a716-446655440001",
```

```
    "spot_price": 415.50,  
    "risk_free_rate": 0.0525,  
    "data_date": "2024-01-15"  
  },  
  {  
    "market_data_id": "660e8400-e29b-41d4-a716-446655440002",  
    "spot_price": 412.30,  
    "risk_free_rate": 0.0520,  
    "data_date": "2024-01-14"  
  }  
],  
"total_records": 10,  
"page": 1  
}
```

### 3. Option Pricing

#### Calculate Option Prices

**POST** /pricing/calculate

#### Request Body:

```
{  
  "option_details": {  
    "symbol": "SPY",  
    "strike_price": 420.00,  
    "expiration_date": "2024-03-15",  
    "option_type": "CALL"  
  },  
  "market_data": {  
    "spot_price": 415.50,  
    "risk_free_rate": 0.0525  
  },  
  "input_parameters": {  
    "volatility": 0.20,  
    "tail_index_alpha": 3.0,  
    "anchor_strike": 410.00,  
    "anchor_price": 8.50  
  }  
}
```

#### Response:

```
{  
  "request_id": "770e8400-e29b-41d4-a716-446655440003",  
  "status": "COMPLETED",  
  "calculation_time_ms": 125,  
  "option_details": {
```

```
    "symbol": "SPY",
    "strike_price": 420.00,
    "expiration_date": "2024-03-15",
    "option_type": "CALL",
    "time_to_expiry": 0.16667
  },
  "input_parameters": {
    "spot_price": 415.50,
    "volatility": 0.20,
    "risk_free_rate": 0.0525,
    "tail_index_alpha": 3.0,
    "anchor_strike": 410.00,
    "anchor_price": 8.50
  },
  "results": {
    "bsm_price": 7.25,
    "pareto_price": 8.75,
    "market_reference_price": 8.50
  },
  "comparisons": {
    "pareto_vs_bsm": {
      "difference": 1.50,
      "percentage": 20.69
    },
    "market_vs_bsm": {
      "difference": 1.25,
      "percentage": 17.24
    },
    "pareto_vs_market": {
      "difference": 0.25,
      "percentage": 2.94
    }
  },
  "created_at": "2024-01-15T14:30:00Z"
}
```

## Batch Calculate Multiple Options

**POST** /pricing/batch-calculate

### Request Body:

```
{
  "requests": [
    {
      "option_details": {
        "symbol": "SPY",
        "strike_price": 420.00,
        "expiration_date": "2024-03-15",
        "option_type": "CALL"
      },

```

```

    "input_parameters": {
      "volatility": 0.20,
      "tail_index_alpha": 3.0,
      "anchor_strike": 410.00,
      "anchor_price": 8.50
    }
  },
  {
    "option_details": {
      "symbol": "SPY",
      "strike_price": 425.00,
      "expiration_date": "2024-03-15",
      "option_type": "CALL"
    },
    "input_parameters": {
      "volatility": 0.22,
      "tail_index_alpha": 3.2,
      "anchor_strike": 415.00,
      "anchor_price": 6.75
    }
  }
],
"market_data": {
  "spot_price": 415.50,
  "risk_free_rate": 0.0525
}
}

```

### Response:

```

{
  "batch_id": "880e8400-e29b-41d4-a716-446655440004",
  "total_requests": 2,
  "completed": 2,
  "failed": 0,
  "results": [
    {
      "request_id": "770e8400-e29b-41d4-a716-446655440005",
      "status": "COMPLETED",
      "option_details": {
        "strike_price": 420.00,
        "option_type": "CALL"
      },
      "results": {
        "bsm_price": 7.25,
        "pareto_price": 8.75,
        "market_reference_price": 8.50
      }
    },
    {
      "request_id": "770e8400-e29b-41d4-a716-446655440006",
      "status": "COMPLETED",

```

```
    "option_details": {
      "strike_price": 425.00,
      "option_type": "CALL"
    },
    "results": {
      "bsm_price": 5.80,
      "pareto_price": 7.15,
      "market_reference_price": 6.75
    }
  }
]
```

## 4. Pricing History

### Get Pricing Request History

**GET** /pricing/history

#### Query Parameters:

- limit** (optional): Number of records (default: 20, max: 100)
- offset** (optional): Pagination offset (default: 0)
- symbol** (optional): Filter by symbol
- start\_date** (optional): YYYY-MM-DD format
- end\_date** (optional): YYYY-MM-DD format

#### Example Request:

```
GET /pricing/history?limit=10&symbol=SPY&start_date=2024-01-10
Authorization: Bearer <jwt_token>
```

#### Response:

```
{
  "total_records": 45,
  "limit": 10,
  "offset": 0,
  "data": [
    {
      "request_id": "770e8400-e29b-41d4-a716-446655440003",
      "option_details": {
        "symbol": "SPY",
        "strike_price": 420.00,
        "expiration_date": "2024-03-15",
        "option_type": "CALL"
      },
      "results": {
        "bsm_price": 7.25,
```

```
    "pareto_price": 8.75,  
    "market_reference_price": 8.50  
  },  
  "comparisons": {  
    "pareto_vs_bsm_percentage": 20.69  
  },  
  "created_at": "2024-01-15T14:30:00Z"  
}  
]  
}
```

## Get Specific Pricing Request

**GET** /pricing/requests/{request\_id}

### Example Request:

```
GET /pricing/requests/770e8400-e29b-41d4-a716-446655440003  
Authorization: Bearer <jwt_token>
```

### Response:

```
{  
  "request_id": "770e8400-e29b-41d4-a716-446655440003",  
  "trader_id": "550e8400-e29b-41d4-a716-446655440000",  
  "status": "COMPLETED",  
  "option_details": {  
    "symbol": "SPY",  
    "strike_price": 420.00,  
    "expiration_date": "2024-03-15",  
    "option_type": "CALL",  
    "time_to_expiry": 0.16667  
  },  
  "input_parameters": {  
    "spot_price": 415.50,  
    "volatility": 0.20,  
    "risk_free_rate": 0.0525,  
    "tail_index_alpha": 3.0,  
    "anchor_strike": 410.00,  
    "anchor_price": 8.50  
  },  
  "results": {  
    "bsm_price": 7.25,  
    "pareto_price": 8.75,  
    "market_reference_price": 8.50  
  },  
  "comparisons": {  
    "pareto_vs_bsm": {  
      "difference": 1.50,  

```



```
    "percentage": 20.69
  },
  "market_vs_bsm": {
    "difference": 1.25,
    "percentage": 17.24
  }
},
"created_at": "2024-01-15T14:30:00Z",
"calculation_time_ms": 125
}
```

## 5. Analytics and Reporting

### Get Pricing Statistics

**GET** /analytics/pricing-stats

#### Query Parameters:

- period** (optional): "1d", "7d", "30d" (default: "7d")
- symbol** (optional): Filter by symbol

#### Example Request:

```
GET /analytics/pricing-stats?period=30d&symbol=SPY
Authorization: Bearer <jwt_token>
```

#### Response:

```
{
  "period": "30d",
  "symbol": "SPY",
  "statistics": {
    "total_requests": 156,
    "average_bsm_price": 12.45,
    "average_pareto_price": 14.78,
    "average_pareto_vs_bsm_percentage": 18.7,
    "price_ranges": {
      "bsm_min": 2.10,
      "bsm_max": 35.60,
      "pareto_min": 2.85,
      "pareto_max": 42.30
    }
  },
  "volatility_stats": {
    "avg_volatility": 0.22,
    "min_volatility": 0.15,
    "max_volatility": 0.35
  },
  "alpha_stats": {
```

```
    "avg_alpha": 3.2,  
    "min_alpha": 2.5,  
    "max_alpha": 4.0  
  }  
},  
"generated_at": "2024-01-15T16:00:00Z"  
}
```

## Get Trader Activity Summary

**GET** /analytics/activity-summary

### Query Parameters:

- period** (optional): "1d", "7d", "30d" (default: "7d")

### Example Request:

```
GET /analytics/activity-summary?period=7d  
Authorization: Bearer <jwt_token>
```

### Response:

```
{  
  "period": "7d",  
  "trader_id": "550e8400-e29b-41d4-a716-446655440000",  
  "summary": {  
    "total_requests": 28,  
    "daily_average": 4.0,  
    "symbols_traded": ["SPY", "QQQ", "IWM"],  
    "option_types": {  
      "CALL": 18,  
      "PUT": 10  
    },  
  },  
  "most_active_day": "2024-01-12",  
  "most_active_day_requests": 8  
},  
  "daily_breakdown": [  
    {  
      "date": "2024-01-15",  
      "requests": 6  
    },  
    {  
      "date": "2024-01-14",  
      "requests": 4  
    }  
  ]  
}
```

# Error Handling

## Standard Error Response Format

```
{
  "error": {
    "code": "VALIDATION_ERROR",
    "message": "Invalid input parameters",
    "details": {
      "field": "volatility",
      "issue": "Value must be between 0.01 and 2.0"
    },
    "timestamp": "2024-01-15T14:30:00Z",
    "request_id": "req_123456789"
  }
}
```

## HTTP Status Codes

- **200 OK** - Successful request
- **201 Created** - Resource created successfully
- **400 Bad Request** - Invalid request data
- **401 Unauthorized** - Invalid or missing authentication
- **403 Forbidden** - Insufficient permissions
- **404 Not Found** - Resource not found
- **422 Unprocessable Entity** - Validation errors
- **429 Too Many Requests** - Rate limit exceeded
- **500 Internal Server Error** - Server error

## Common Error Codes

- **INVALID\_TOKEN** - JWT token is invalid or expired
- **VALIDATION\_ERROR** - Request data validation failed
- **CALCULATION\_ERROR** - Error in option pricing calculation
- **RATE\_LIMIT\_EXCEEDED** - Too many requests
- **SYMBOL\_NOT\_FOUND** - Market data not available for symbol
- **INSUFFICIENT\_PERMISSIONS** - User lacks required permissions

## Rate Limiting

- **Rate Limit:** 100 requests per minute per trader
- **Burst Limit:** 20 requests per 10 seconds
- **Headers Returned:**

```
X-RateLimit-Limit: 100
X-RateLimit-Remaining: 95
X-RateLimit-Reset: 1642262400
```

## Pagination

For endpoints returning lists, pagination follows this pattern:

- **limit**: Number of items per page (max 100)
- **offset**: Number of items to skip
- Response includes **total\_records**, **limit**, and **offset**

## WebSocket API (Real-time Updates)

### Connection

```
wss://api.optionpricing.com/v1/ws?token=<jwt_token>
```

### Subscribe to Real-time Market Data

```
{
  "action": "subscribe",
  "type": "market_data",
  "symbols": ["SPY", "QQQ"]
}
```

### Real-time Market Data Update

```
{
  "type": "market_data_update",
  "data": {
    "symbol": "SPY",
    "spot_price": 416.20,
    "timestamp": "2024-01-15T14:35:00Z"
  }
}
```

## API Versioning

- Current version: **v1**
- Version specified in URL path: **/v1/...**
- Backward compatibility maintained for at least 12 months
- Deprecation notices provided 6 months in advance

## Security Considerations

- All endpoints require HTTPS in production
- JWT tokens expire after 1 hour

- Rate limiting to prevent abuse
- Input validation on all parameters
- SQL injection prevention
- CORS configuration for web clients
- API key rotation support for service accounts